

Next.js Technical Implementation Blueprint

A source-grounded specification for rebuilding the supplied Business Central extension as a secure, modular web application.

Source assessed	Implementation target	Document status
Nation CBS AL extension (v1.0.0.0); 665 AL files	Next.js App Router + TypeScript + relational database	Build blueprint - assumptions identified where source is not definitive

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1. Executive summary

This is a techno-functional implementation guide for a Next.js product. It organizes the system by business module, user outcome, workflow, data, controls, and delivery acceptance - not by source-code conversion.

What was found. Nation CBS is a SACCO / cooperative financial-services extension built on Microsoft Dynamics 365 Business Central. Its implementation is composed of 163 custom tables, 268 pages, 98 reports, 28 codeunits, 40 enums, 7 XML imports, and extensions to base ERP entities. It depends on a separate Nation ERP extension and Business Central platform services.

Recommended product boundary. Rebuild it as a modular core-banking operations portal. Keep financial posting and Business Central-specific ledger behavior behind explicit domain services. Treat the first release as an operational web application with auditable workflows, role-based access, imports, document handling, and integration adapters.

Important limitation. The source shows user-facing pages, table schemas, reports, and business procedures. It does not supply a complete external API contract, production configuration values, authentication design, reporting layouts, or runtime data. Those must be confirmed in discovery before production cutover.

2. Source inventory and interpretation

AL asset	Count	Next.js interpretation
Custom tables	163	Persistent domain entities and operational ledgers
Pages	268	List, detail, wizard, dashboard, and setup screens
Reports	98	Operational reports, statements, certificates, schedules, notices
Codeunits	28	Domain services, approvals, posting routines, integrations, scheduled operations
Enums / extensions	43	Controlled vocabularies and ERP type extensions
XMLports	7	Bulk import pipelines: checkoff, SMS, dividends, debts/loans
Queries	4	Read-model / reporting queries
App dependency	1	Nation ERP v1.0.0.0; requires replacement interfaces or retained ERP integration

The most important codeunits are Product Management, Member Management, Loans Management, Payroll Loan Management, Fixed Deposit Management, Journal Management, Checkoff Management, Scheduled Activities, Notifications Management, Integrations Management, Channels Integrations, FOSA Management, Cash Management, Share Trading Management, Dividend Management, ATM Integration, and Custodial Management.

3. Target architecture

Use a modular monolith first. It matches the source's transactional coupling, allows a single financial audit trail, and keeps deployment simple. Extract adapters or workers only for unreliable / asynchronous external systems.

Browser UI -> Next.js App Router -> Server Actions / Route Handlers -> Domain services -> PostgreSQL
SharePoint, payroll, channels -> Outbox + worker -> SMS, payments, ATM,
generated reports -> Object storage -> attachments and

Layer	Recommended choice	Responsibility
Web	Next.js 15+ App Router, TypeScript, React Server Components	Authenticated operations UI, forms, dashboards, document views
API	Route Handlers + typed service layer	Internal UI API; partner webhooks; OpenAPI for public integrations
Persistence	PostgreSQL + Prisma or Drizzle	Transactions, audit records, immutable posting journals, row-level constraints
Auth	OIDC provider (Auth.js / enterprise IdP)	SSO, MFA, session policy, role and branch scoping
Async	BullMQ + Redis or managed queue	Imports, scheduled jobs, report runs, retries, notifications, integration outbox
Storage	S3-compatible object storage	Attachments, applications, statement PDFs, export files
Observability	Structured logs, error tracking, metrics	Correlation IDs, posting traceability, job failure alerts

Rule: no direct database mutation from UI components. Every monetary or approval state transition must call a server-side domain command inside a database transaction.

4. Module map and navigation

The source's Role Center exposes broad operational navigation. Use route groups by business capability, and use list/detail patterns consistently.

Module	Source evidence	Suggested routes / responsibilities
Identity & admin	Role Center; User Profile Management; User Setup extension	/admin/users, /roles, /branches, /settings, /audit
Member lifecycle	Members, Member Application, KINs, accounts, activation, withdrawal	/members, /member-applications, /members/[id], /withdrawals
Products & accounts	Sacco Products, categories, charges, member accounts, fixed deposits	/products, /accounts, /fixed-deposits
Loans & securities	Loans, schedules, guarantees, collateral, moratorium, recovery	/loans, /loan-applications, /collateral, /guarantees, /recoveries
Cash / FOSA / teller	FOSA transactions, teller transactions, denomination, cash management	/teller, /cash, /fosa, /journals
Payroll & checkoff	Employers, payroll codes, uploads, calculations, advice	/employers, /checkoffs, /payroll
Dividends & shares	Dividend headers/lines/recoveries, share trading/floating/transfers	/dividends, /shares, /share-transfers
Payments & channels	Standing orders, PesaLink, B2B, mobile, ATM, channel transactions	/payments, /standing-orders, /channels, /atm
Compliance & custodial	AML checks, custodial services, document checklist	/compliance, /custodial, /documents
Reports	98 SSRS/RDLC report assets	/reports, /statements, /exports

The counts in the source indicate especially large Member (49 pages), Loan (41), Share (13), Dividend (11), Channel (10), Checkoff (9), ATM (8), and Mobile (8) surface areas. Build the shared patterns first: filters, status chips, work queues, detail tabs, attachments, approval actions, CSV import/export, and printable documents.

5. Module implementation cards

Use each card as a build backlog and UAT checklist. A module is complete only when users can execute its happy path, approval path, exception path, audit trace, and core output.

5.1 Member management and onboarding

Purpose and actors	Records and screens	Lifecycle, controls and acceptance
Register, verify, activate, maintain, and exit individual or corporate members, including their default accounts and KYC evidence. Actors: Member services, KYC/compliance, branch staff, approvers.	Records: Member Application, Member, Member Edit, Member Account, Category, Nominee/KIN, Activation, Withdrawal, Document. Screens: /member-applications, /members, /members/[id], /members/[id]/accounts, /members/[id]/documents, /withdrawals.	Lifecycle: Draft -> validate/KYC -> approval -> member + accounts created -> active; edits and withdrawals use separate approvals. Controls: Validate identity, KRA PIN, phone, age/date and duplicate rules; category determines default accounts; restrict/document member views; no withdrawal until obligations settle. Outputs: Membership form/list, KIN list, statement, withdrawal documents. Done: A user completes individual and corporate onboarding, approvals cannot be bypassed, attachments are controlled, and resulting accounts reconcile to opening entries.

5.2 Products, accounts and fixed deposits

Purpose and actors	Records and screens	Lifecycle, controls and acceptance
Configure member financial products, pricing, charges, account mappings, and deposit lifecycle. Actors: Product administrators, finance, branch operations.	Records: Product, Category, Interest Band, Charge Setup, Member Account, Fixed Deposit Type/Account/Schedule. Screens: /products, /products/[code], /charges, /accounts, /fixed-deposits.	Lifecycle: Setup -> approved/active product -> account opening -> transaction/accrual -> maturity, renewal or closure. Controls: Require posting type, account mapping, rates and charge policy before activation; prevent transactions on blocked products; selected interest band must cover the term. Outputs: Account list, fixed-deposit certificate, maturity/accrual reports. Done: Configured products can open compliant accounts and produce correct schedule, maturity, and certificate outcomes.

5.3 Loans, appraisal, security and recovery

Purpose and actors	Records and screens	Lifecycle, controls and acceptance
Originate, appraise, approve, disburse, service, restructure, secure, and recover member loans. Actors: Credit officers, guarantors, approvers, collections, finance.	Records: Loan, Schedule, Security, Guarantee, Collateral, Appraisal, Interest Accrual, Moratorium, Recovery, Defaulter Notice. Screens: /loan-applications, /loans, /loans/[id], /guarantees, /collateral, /recoveries, /moratoriums.	Lifecycle: Application -> appraisal -> security -> approval -> disbursement/posting -> repayment/accrual -> arrears/recovery -> close or controlled restructure. Controls: Enforce eligibility, contributions, product terms, charges, security limits, mandatory documents, maker-checker and controlled reversal rather than posted edits. Outputs: Application/appraisal, repayment schedule, register, ageing, disbursement, defaulter and guarantor reports. Done: Signed-off sample cases reproduce correct schedule, journal, repayment allocation, arrears classification and recovery result.

5.4 Teller, FOSA, cash and journals

Purpose and actors	Records and screens	Lifecycle, controls and acceptance
Run member cash services and internal financial movements with daily till and journal control. Actors: Tellers, supervisors, cashiers, finance.	Records: Teller Setup/Transaction, FOSA Transaction, Denomination, Journal Voucher, Cheque Deposit, Bankers Cheque. Screens: /teller, /fosa, /cash, /journals, /receipts, /payments, /cheque-deposits.	Lifecycle: Initiate -> validate balance/account -> approve if needed -> post -> receipt/journal -> reconcile. Cheques retain clearing states. Controls: Enforce till entitlement and denomination balance; posting setup complete; debit = credit; immutable entries and reversal-only correction. Outputs: Cash receipt, cash book, deposit slip, vouchers, bank balance report. Done: Deposit, withdrawal, end-of-day reconciliation and trace to approval/journal all work.

5.5 Payroll and checkoff collections

Purpose and actors	Records and screens	Lifecycle, controls and acceptance
Receive employer deductions, validate them, allocate member obligations, and issue advice/exceptions. Actors: Payroll officers, employer liaison, collections, finance.	Records: Employer, Payroll Code, Checkoff Upload/Header/Line/Calculation, Variation, Advice. Screens: /employers, /payroll, /checkoffs, /checkoffs/imports, /checkoffs/[batch].	Lifecycle: Upload -> stage/preview -> correct rows -> approve -> post allocation -> advice/reconcile. Controls: Preserve raw file/errors; no direct upload posting; prevent duplicate period batch; match member/account; reconcile batch total to allocations. Outputs: Checkoff advice, monthly receipts, variance, under/over-paid analysis. Done: Representative payroll has row-level errors, correction, one-time approval/posting and full employer-total reconciliation.

5.6 Dividends, shares and member capital

Purpose and actors	Records and screens	Lifecycle, controls and acceptance
Calculate and distribute dividends, recover obligations, and manage share activity. Actors: Finance, member services, dividend committee, approvers.	Records: Dividend Header/Line/Recoveries/Entries, Share Trading, Floating, Transfer. Screens: /dividends, /dividends/[period], /share-trading, /share-floating, /share-transfers.	Lifecycle: Freeze parameters -> calculate -> review/approve -> recover/post -> issue result. Share actions have independent approvals. Controls: Freeze balance date/input; retain calculation evidence; no duplicate period posting; distinguish calculated, recovered and payable values. Outputs: Dividend slip, transfer receipt, share detail and classification reports. Done: A signed-off sample run is reproducible, traceable at member level and posts once.

5.7 Payments, standing orders and channels

Purpose and actors	Records and screens	Lifecycle, controls and acceptance
Process payment instructions and external-channel activity through settlement and reconciliation. Actors: Payments operations, channel support, finance.	Records: Standing Order, PesaLink, B2B, Channel Transaction/Setup/Dump/Reversal, Account Instruction. Screens: /payments, /standing-orders, /pesalink, /b2b, /channels.	Lifecycle: Create -> approve -> submit -> callback/poll -> settle/post -> archive/reconcile; failed/skipped/reversed items remain actionable. Controls: Unique external reference/idempotency; signed webhooks; full status history; never treat timeout as failure; daily settlement reconciliation. Outputs: Standing order register, payment due, repayment schedule, exception queues. Done: Delayed/duplicate callbacks never double-post, and each external reference reconciles to settlement.

5.8 Mobile, ATM and communications

Purpose and actors	Records and screens	Lifecycle, controls and acceptance
Manage channel enrolment/cards, ingest activity/reversals, and send member communications securely. Actors: Digital operations, card team, support, member services.	Records: Mobile Application/Member/Ledger, ATM Application/Card/Transaction, SMS Ledger, Bulk SMS. Screens: /mobile, /mobile-applications, /atm, /atm-cards, /atm-transactions, /messages.	Lifecycle: Application -> approve -> link -> active; receive transaction -> validate -> post/reverse -> notify. Bulk SMS: audience -> queue -> delivery outcome. Controls: Mask card values; validate card/member/account link; retain reversal reference; separate received from posted; apply consent/opt-out and secure logs. Outputs: ATM/mobile eligibility and transaction reports; message delivery/failure log. Done: Sandbox enrolment, posting, reversal, masking and retry behavior are demonstrated.

5.9 Compliance, documents and custodial

Purpose and actors	Records and screens	Lifecycle, controls and acceptance
Control KYC evidence, AML review, document checklists, custodial holdings and liens. Actors: Compliance, custodial operations, auditors, approvers.	Records: Attachment/Checklist, AML Check, Custodial Header/Movement/Service/Entry, Lien, View Log. Screens: /compliance, /aml-checks, /documents, /custodial, /liens, /audit.	Lifecycle: Evidence -> checklist/review -> approval -> active exception/service -> monitored change -> closure/release. Controls: Role-limited document access; checksum/version/metadata; lien and custodial movement require reason + approval; views are audited. Outputs: Custodial receipt/holding, collateral/lien, AML and document-completeness queues. Done: Auditor can trace regulated action to evidence, approval, access history and immutable movement/posting.

5.10 Administration, reports and operations control

Purpose and actors	Records and screens	Lifecycle, controls and acceptance
Manage master setup, permissions, scheduled processing, work queues, reports, and support visibility. Actors: Administrators, branch managers, finance, operations, auditors.	Records: Sacco Setup, User Setup, Lookup Values, Scheduled Activities, Job Execution, Cues, Report Runs. Screens: /admin/settings, /admin/users, /admin/jobs, /dashboards, /reports, /audit.	Lifecycle: Configure -> validate -> activate; job -> record result -> retry/escalate; dashboards surface pending work. Controls: Audit configuration changes; secrets in vault; job locking/idempotency; scoped report export; retain parameters and source cutoff. Outputs: Operational dashboards plus member, loan, collections, cash, dividend and channel report runs. Done: Admin config, controlled job retry, role-scoped report and traceable run history all operate reliably.

6. Cross-module data model

Model the source records as normalized entities plus an append-only accounting and audit layer. Do not recreate Business Central FlowFields as stored mutable totals; calculate them from ledger entries or materialized projections.

Aggregate	Core entities	Key relationships / constraints
Member	Member, MemberApplication, MemberAccount, NomineeKin, MemberDocument	Member has many accounts, applications, KINs, documents; identity number unique per identity type
Product	Product, ProductCategory, ProductCharge, InterestBand	Product controls posting type, rates, account mappings, eligibility, charges
Loan	Loan, LoanSchedule, LoanSecurity, Guarantee, Collateral, LoanRecovery	Loan belongs to member/product; schedules versioned; guarantee/collateral allocations cannot exceed eligible balance
Money movement	Transaction, TransactionLine, Account, Journal, Settlement	Double-entry immutable posted entries; idempotency key on inbound integrations
Workflow	ApprovalRequest, ApprovalStep, ApprovalDecision, Comment	Document type + ID; transitions only through policy service
Payroll	Employer, PayrollCode, CheckoffBatch, CheckoffLine	Batch import retains raw file, validation result, row errors, and posting reference
Channels	ChannelTransaction, ATMCARD, MobileMember, StandingOrder	External reference unique per channel; status/outcome and retry history retained
Documents	Attachment, DocumentChecklist, GeneratedDocument	Metadata in DB, binary in object storage, virus scan status

Recommended conventions: UUID primary keys externally; human-readable business numbers from a sequencer; ISO dates in UTC timestamps; Decimal / NUMERIC for money; currency code on every monetary value; soft-delete only for reference data; never delete posted financial entries.

6.1 Core data dictionary - members, products and loans

Table	Key fields and recommended PostgreSQL type	Relationships / implementation notes
members	id UUID PK; member_no VARCHAR(20) UNIQUE; first_name/middle_name/last_name VARCHAR(50); full_name VARCHAR(150); mobile_phone VARCHAR(50); identification_no VARCHAR(50); date_of_birth DATE; status member_status ENUM; created_at TIMESTAMPTZ	Source: Members. Add identity_type and unique (identity_type, identification_no). Encrypt / mask sensitive identity fields; retain audit/version history.
products	id UUID PK; code VARCHAR(20) UNIQUE; category_code VARCHAR(20); description VARCHAR(100); posting_type product_posting_type ENUM; interest_rate NUMERIC(19,4); interest_method loan_rate_type ENUM; active BOOLEAN	Source: Sacco Products. Do not store chart-account codes without a validated account mapping table. Product changes need effective dating.
loans	id UUID PK; loan_no VARCHAR(20) UNIQUE; member_id UUID FK; product_id UUID FK; requested_amount/approved_amount/disbursed_amount NUMERIC(19,4); interest_rate NUMERIC(19,4); installments INTEGER; application_date DATE; repayment_start_date DATE; status document_status ENUM	Source: Loans. Maintain status and posted_at separately; monetary values carry currency; enforce non-negative and approved <= permitted policy.
loan_schedules	id UUID PK; loan_id UUID FK; installment_no INTEGER; expected_date DATE; principal_repayment NUMERIC(19,4); interest_repayment NUMERIC(19,4); total_repayment NUMERIC(19,4); running_balance NUMERIC(19,4)	Source: Loan Schedule. Unique (loan_id, installment_no); version schedules when restructured; totals reconcile to approved principal and calculated interest.

6.2 Core data dictionary - security, collections and distributions

Table	Key fields and recommended PostgreSQL type	Relationships / implementation notes
loan_guarantees	id UUID PK; loan_id UUID FK; guarantor_member_id UUID FK; guaranteed_amount NUMERIC(19,4); available_guarantee NUMERIC(19,4); self_guarantee BOOLEAN; substitution_status ENUM	Source: Loan Guarantees. Available value is derived / locked during approval; prevent total allocations above permitted member guarantee.
collateral	id UUID PK; collateral_no VARCHAR(20) UNIQUE; member_id UUID FK; collateral_type VARCHAR(20); description VARCHAR(150); collateral_value NUMERIC(19,4); status collateral_status ENUM; serial_reg_no VARCHAR(100); insurance_expiry DATE	Source: Collateral Register. Link collateral allocations to loans in a child table; retain valuation evidence and release authorization.
checkoff_batches	id UUID PK; batch_no VARCHAR(20) UNIQUE; employer_id UUID FK; document_date DATE; posting_date DATE; uploaded_amount NUMERIC(19,4); status document_status ENUM; posted BOOLEAN; source_file_id UUID FK	Source: Checkoff Header. Unique (employer_id, payroll_period, upload_type) after confirming business rule. Preserve file checksum, parser version and validation state.
checkoff_lines	id UUID PK; batch_id UUID FK; member_id UUID FK NULL; payroll_no VARCHAR(20); amount_earned NUMERIC(19,4); recoveries NUMERIC(19,4); net_amount NUMERIC(19,4); posted BOOLEAN; row_status import_row_status ENUM	Source: Checkoff Lines. Support unmatched / suspense rows. Keep original row payload and error messages for remediation.
dividend_runs	id UUID PK; run_no VARCHAR(20) UNIQUE; start_date/end_date DATE; transaction_code VARCHAR(20); expense_account_id UUID; payable_account_id UUID; recover_loans BOOLEAN; status dividend_status ENUM	Source: Dividend Header. Freeze parameter snapshot and input cutoff; a run is immutable once posted.
dividend_lines	id UUID PK; dividend_run_id UUID FK; member_id UUID FK; automatic_amount NUMERIC(19,4); manual_amount NUMERIC(19,4); recoveries NUMERIC(19,4); net_amount NUMERIC(19,4); posted BOOLEAN; notified BOOLEAN	Source: Dividend Lines. Unique (dividend_run_id, member_id); track manual override reason and approver.

6.3 Core data dictionary - transactions, channels and controls

Table	Key fields and recommended PostgreSQL type	Relationships / implementation notes
teller_transactions	id UUID PK; document_no VARCHAR(20) UNIQUE; transaction_type ENUM; member_id UUID FK; account_id UUID FK; amount NUMERIC(19,4); teller_user_id UUID; till_id UUID; status document_status ENUM; posted BOOLEAN	Source: Teller Transactions. Require a separate till/session table and a link to the final accounting transaction.
fosa_transactions	id UUID PK; document_no VARCHAR(20) UNIQUE; transaction_type fosa_transaction_type ENUM; source_account_id UUID; destination_account_id UUID; amount NUMERIC(19,4); denominations NUMERIC(19,4); status document_status ENUM; posted BOOLEAN	Source: FOSA Transactions. Validate source/destination difference and transaction-specific account rules.
channel_transactions	id UUID PK; channel ENUM; external_reference VARCHAR(250); direction ENUM; transaction_type VARCHAR(20); account_id UUID; member_id UUID; amount NUMERIC(19,4); status channel_status ENUM; received_at TIMESTAMPTZ; posted_at TIMESTAMPTZ	Source: Channel Transactions and ATM Transactions. Unique (channel, external_reference); raw callback in separate immutable channel_messages table.
atm_cards	id UUID PK; card_token VARCHAR(100) UNIQUE; member_id UUID FK; account_id UUID FK; atm_type_id UUID FK; status card_status ENUM; issued_at DATE; collected_at DATE	Source: ATM Cards/Applications. Never store full PAN unless contract and PCI scope explicitly allow it; use token/masked display.
accounting_transactions	id UUID PK; transaction_no VARCHAR(30) UNIQUE; source_type VARCHAR(50); source_id UUID; posting_date DATE; status posting_status ENUM; idempotency_key VARCHAR(100) UNIQUE; posted_at TIMESTAMPTZ	New core ledger boundary. Child accounting_transaction_lines store account_id, debit NUMERIC(19,4), credit NUMERIC(19,4), currency_code and dimensions. Enforce balanced entries.
approval_requests	id UUID PK; document_type VARCHAR(50); document_id UUID; state approval_state ENUM; submitted_by UUID; current_step INTEGER; amount NUMERIC(19,4); decided_at TIMESTAMPTZ	New workflow boundary based on AL approval services. Child decisions include approver, decision, comment, timestamp and delegated authority evidence.
attachments	id UUID PK; owner_type VARCHAR(50); owner_id UUID; file_name VARCHAR(255); storage_key VARCHAR(500); checksum CHAR(64); content_type VARCHAR(100); scan_status ENUM; uploaded_by UUID; uploaded_at TIMESTAMPTZ	Source: document attachment service. Binary remains in object storage; use signed access and a document-access audit log.

7. Workflow and financial-posting design

The source contains explicit send/cancel/release/reopen approval handlers for at least 30 document types, including loan applications/disbursements/restructures, collateral, member applications and edits, teller transactions, liens, standing orders, fixed deposits, ATM and mobile applications, checkoff, member withdrawal, dividend headers, FOSA transactions, cheque deposits, AML checks, and share floating.

Draft -> Pending approval -> Approved -> Posted / Processed
-> Rejected -> Reopened -> Draft

Implement as a state machine. Maintain a document-type configuration table with allowed states, role/group policy, amount thresholds, required checklist items, and whether a posted reversal is required. Every transition writes an immutable ApprovalDecision and AuditEvent. Approval comments must be mandatory on rejection, matching the source's rejection-comment check.

Posting. A command such as `postLoanDisbursement()` must validate the approved state, lock its aggregate row, derive accounting lines, write one balanced Transaction and child lines, mark the document posted, create an outbox event, and commit atomically. Retrying the same command must return the original result via a unique idempotency key.

8. API and command design

Pattern	Examples	Notes
Reads	GET /api/members?status=active; GET /api/loans/:id	Cursor pagination, filter schema, field-level permissions
Commands	POST /api/loans; POST /api/loans/:id/submit; POST /api/loans/:id/post	Separate commands from resource updates
Approvals	POST /api/approvals/:id/approve; POST /api/approvals/:id/reject	Require decision reason / comment as policy dictates
Imports	POST /api/checkoffs/import; GET /api/imports/:id	Async parse, preview, validation, then explicit commit
Webhooks	POST /api/webhooks/atm; /mobile; /payments	Signature verification, raw payload persistence, idempotency
Reports	POST /api/reports/:name/runs; GET /api/report-runs/:id	Async render; signed short-lived file URL

Use Zod schemas shared by client and server. Publish an OpenAPI contract for partners, but never expose internal posting endpoints directly without strong client authentication, scoped authorization, replay protection, and rate limits.

9. Integration blueprint

The source identifies multiple external touchpoints. Implement each as a replaceable adapter behind a stable application service; do not put partner-specific payload logic in route handlers.

Adapter	Source signal	Required behavior
SMS / notifications	Notifications Management, Bulk SMS, SMS URL configuration	Template messages; queued delivery; provider callback; retry / dead-letter; opt-out and PII-safe logs
Document storage	Document Attachment Management: SharePoint upload/link	Presigned upload; malware scan; checksum; metadata; optional SharePoint adapter
ATM	ATM Integration, cards, ATM transactions, reversals	Inbound webhook; card masking; settlement mapping; reversal lifecycle; reconciliation
Mobile / channel	Mobile Application, Mobile Transactions, Channel Transactions	Signed callbacks; external reference idempotency; status polling / webhook recovery
PesaLink / B2B	PesaLink, Coop B2B Integration, B2B Transactions	Payment initiation, callback, settlement and archive/reconciliation states
Payroll / checkoff	XML imports, Checkoff Management, employers	File staging, field mapping, row-level error file, approval then posting
Identity / compliance	Member Management IPRS/KRA validations; AML Management	Secure API vault, consent / legal basis, audit and response retention

Use an outbox table written in the same transaction as each business event. A worker publishes events and records delivery attempts. This avoids the common failure mode where a transaction is posted but the SMS, external channel update, or generated document is silently lost.

10. Security, control, and audit requirements

- Adopt least-privilege RBAC with permissions at module, action, field, and branch/employer scope. Model Business Central user setup rules as policy records, not scattered UI checks.
- Require MFA through the identity provider. Use short sessions, CSRF protection for cookie flows, secure headers, and request/response validation.
- Encrypt sensitive fields and object-store payloads at rest. Redact national IDs, KRA PINs, card numbers, phones, and account numbers from logs. The source includes card masking and identity validation behavior; retain both.
- Store audit events for create/update/state transition/post/reverse/export/view. Capture actor, role, timestamp, correlation ID, old/new values with sensitive-field masking, and reason.
- Separate duties: initiator cannot approve own financial document; configurable maker-checker thresholds; require comment on rejection; stop mutable updates after posting except controlled reversals.
- Reconcile external payments, ATM activity, and channel transactions daily. Alert on unmatched, duplicated, failed, skipped, or late records.

Obtain a formal data-protection and retention review for the jurisdiction before deployment. The source is financial-services software and handles high-sensitivity personal and financial data.

11. Reporting, documents, and dashboards

There are 98 SSRS/RDLC report definitions in the source. Prioritize outputs that drive operations or compliance: member statement, loan application/appraisal/schedule/register, disbursement schedule/summary, loan balances and ageing, defaulter notices, checkoff advice, cash book, bank balances, receipts, fixed-deposit certificate, collateral register, dividend slip, and share transfer documents.

Build reports from query-specific read models. A report run must persist its parameters, requesting user, version, generated timestamp, source-data cutoff, checksum, and output URI. Render PDFs asynchronously, not in the request thread. For statements and notices, snapshot the source data used to produce the document.

Dashboard	Primary metrics	Drill-down
Operations	pending approvals, unposted imports, failed jobs, cash position	approval request, import batch, job log
Credit	applications, disbursement pipeline, arrears, default bands	loan and member
Members	new/active/withdrawn, account openings, KYC exceptions	member profile
Collections	checkoff received/posted/errors, recovery performance	employer and batch
Channels	pending/failed/reversed transactions, reconciliation aged items	external reference

12. Delivery plan and acceptance gates

Phase	Scope	Exit criteria
0. Discovery (2-4 weeks)	Validate business rules, ledger mapping, reports, partner contracts, data quality and roles	Signed domain glossary, migration mapping, API list, priority report list
1. Platform foundation	Auth/RBAC, audit/outbox, attachments, shared UI, configuration, CI/CD	Threat model passed; audit and role tests; deployment pipeline working
2. Core operations	Members, products/accounts, loan application + approval, basic reporting	Parallel UAT of member / loan workflows; reconciliation rules accepted
3. Financial operations	Posting ledger, FOSA/teller, checkoff/payroll, fixed deposits, dividends/shares	Balanced posting tests; import controls; operational sign-off
4. Channels and integrations	ATM, mobile, payments, SMS, SharePoint, reconciliation	Sandbox and failure/retry tests; webhook idempotency verified
5. Migration and rollout	Historical data, report parity, staff training, cutover support	Trial migrations, count/balance reconciliation, rollback plan, go-live approval

Recommended first vertical slice: Member onboarding -> approval -> member account opening -> document attachment -> audit trail -> member statement. It establishes identity, workflow, documents, account data, reporting, and the common UI patterns before the higher-risk loan and cash modules.

Do not estimate solely from page count. The risk is concentrated in posting, approval, calculation, migration, integration, and report parity. Confirm these before committing a fixed delivery date.

13. Test strategy and migration

- Unit-test calculation, eligibility, schedule, charge, allocation, status-transition, and authorization rules. Golden test cases must be extracted from current production scenarios with anonymized values.
- Use integration tests against mocked SMS/payment/ATM/SharePoint adapters and contract tests against partner sandbox APIs. Verify duplicate callbacks and delayed/reordered messages.
- Use end-to-end tests for maker-checker workflows, import preview/commit, posting/reversal, attachment access, report generation, and permission boundaries.
- Perform financial invariant tests: every posted transaction balances; no unauthorized transition occurs; scheduled totals reconcile; guarantees/collateral are not over-allocated; reversal refers to original transaction.
- Migrate in stages: reference data -> members/KYC -> accounts and opening balances -> loans/schedules/securities -> open workflows -> attachments -> historical reporting. Store source record identifiers for traceability.
- Run at least two trial migrations. Reconcile entity counts, balances by product/member/GL mapping, open loan schedules, and a representative set of statements and reports before cutover.

14. Decisions required before implementation

Decision	Why it matters	Owner
System of record	Whether Next.js replaces or integrates with Business Central changes the ledger and migration design	Business + architecture
Financial posting model	Define chart mapping, reversals, periods, approval gates, and reconciliation	Finance + engineering
Priority scope	The full 268-page surface should be phased, not built as one release	Product + operations
Partner contracts	ATM, mobile, PesaLink/B2B, SMS, identity and SharePoint APIs need validated contracts	Integration owners
Data retention & privacy	Determines encryption, access, archival, consent, and audit requirements	Compliance + legal
Report parity	Choose statutory/operational reports required at go-live and their acceptance samples	Operations + finance

Appendix - source anchors

Key source artifacts assessed: app.json; codeunits 52204000-52204029; tables 52204000-52204208; Role Center page 52204000; 98 SSRS/RDLC report definitions; 7 XMLports for debt/loan, dividend, bulk SMS and checkoff flows. Version and dependency claims are taken from app.json. Module and workflow claims are taken from object names and visible AL procedures.

This blueprint intentionally avoids copying the extension code. It specifies an implementation architecture and transition path suitable for a modern web application.